Gender Criteria Gap in Evaluation: Role of Perceived Intentions and Outcomes

Nisvan Erkal
Lata Gangadharan
Boon Han Koh

[ONLINE APPENDIX]

Appendix A Experimental Instructions

This appendix includes the instructions for the experiment reported in the paper. It also includes the practice questions for the leadership task, and the verbal instructions read out by the experimenter between Stage 1 and Stage 2 of the leadership task.

Overview of Experiment

Thank you for agreeing to take part in this study which is funded by the Australian Research Council. Please read the following instructions carefully. A clear understanding of the instructions will increase your earnings from the experiment.

There are two parts in today's experiment: Part 1 and Part 2. We have provided you with instructions for Part 1, and we will explain them in greater detail shortly. We will hand out instructions for Part 2 at the end of Part 1. At the end of Part 2, you will be asked to complete a post-experimental questionnaire. Please be assured that all your responses and decisions will remain anonymous.

You will be paid for either Part 1 or Part 2 of today's experiment. Hence, you should carefully consider all the decisions you make in today's experiment as they may determine your earnings. Whether you will be paid for Part 1 or Part 2 will be randomly determined at the end of today's session. You will be informed of the outcome of the experiment at the end of the session.

During the experiment, we will be using Experimental Currency Units (ECU). At the end of the session, we will convert the amount you earn into Australian Dollars (AUD) using the following conversion rate: 10 ECU = 1 AUD. You have already earned 50 ECU for completing the pre-experimental questionnaire.

Please do not talk to one another during the experiment, and please refrain from using your mobile phones and/or tablets. We require you to pay attention to the computer screen at all times. If anyone is found using their mobile phones and/or tablets, they may be asked to leave the experiment and may be excluded from future experiments. If you have any questions, please raise your hand and we will come over to answer your questions privately.

Do not turn over to the next page until you have been instructed by the experimenter to do so.

Part 1

You will participate in Part 1 in groups of three. There are two possible roles: Leader and Member. Each group will consist of one Leader and two Members.

Part 1 consists of two stages. In each stage, you will be asked to make decisions relating to five investment tasks. The following two sections explain the decisions that you will make in each stage.

(i) Stage 1: Investment decisions as Leaders

In Stage 1, you will be asked to make a decision for all five investment tasks assuming that you are the Leader of your group.

You will be informed whether you are the Leader at the end of Stage 1. Your decisions will be implemented if you are assigned to be the Leader of your group.

For each investment task, you will be given an endowment of 300 ECU. You will be asked to choose between two investment options. Your choice will affect both your payoff and your Members' payoffs. Each investment can either fail or succeed. The two investment options have different chances of success/failure, as well as different costs to you.

Specifically, the two investments are:

> **Investment X:** This investment will succeed with a 75% chance and fail with a 25% chance, and it costs you 200 ECU.
>
> **Investment Y:** This investment will succeed with a 25% chance and fail with a 75% chance, and it costs you 50 ECU.

Each investment provides you and your Members a high return if it succeeds, and a low return if it fails.

Your payoff and your Members' payoffs are calculated as follows:

> Payoff to you (Leader) = 300 ECU – Cost of investment + Returns on investment
>
> Payoff to each Member = Returns on investment

Note that the returns of the two investments may be different for each task, and this will therefore affect the final payoffs to you and your Members. However, the investments always provide a higher return if they succeed and a lower return if they fail. Please pay attention to these numbers on the screen for each task.

Figure 1 shows an example where Investment X and Investment Y provide you and each Member a return of 275 ECU if they succeed and 50 ECU if they fail. These numbers are shown in red.

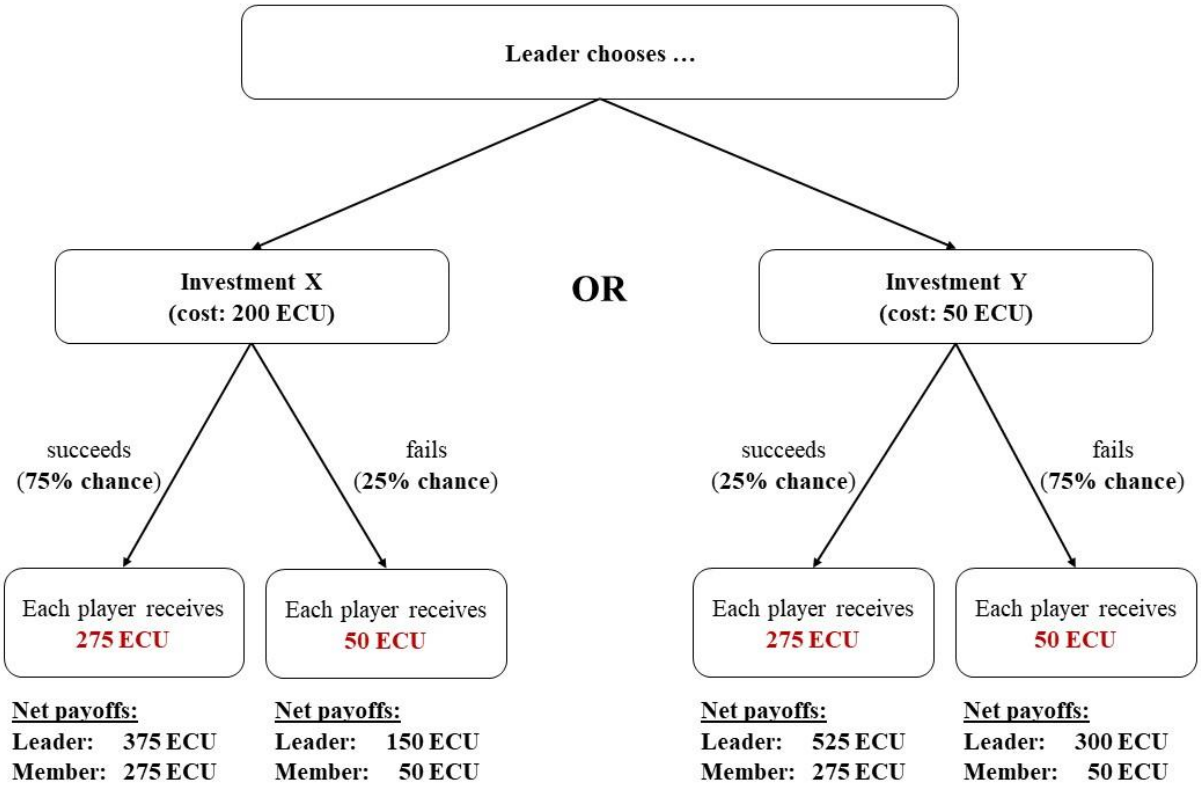


Figure 1: Investment Options in Part 1 (Example of a Task)

Example 1. Suppose in the task depicted in Figure 1, you choose Investment X. Then, the investment costs you 200 ECU, and will succeed with a 75% chance and fail with a 25% chance. If the investment succeeds, then you will receive (300 – 200 + 275) = 375 ECU and each Member will receive 275 ECU.

Example 2. Suppose in the task depicted in Figure 1, you choose Investment Y. Then, the investment costs you 50 ECU, and will succeed with a 25% chance and fail with a 75% chance. If the investment fails, then you will receive (300 – 50 + 50) = 300 ECU and each Member will receive 50 ECU.

The other Members of your group will never learn your investment decisions. At the end of the experiment, they will learn how much they have received from the chosen investment, but they will not learn your investment decision.

(ii) Stage 2: Members' predictions and decisions

At the beginning of Stage 2, you will be provided information about your groups and roles. Hence, you will be informed whether you are the Leader or a Member after you have completed Stage 1, and before Stage 2 begins.

Predictions of Leader's decisions: As a Member, you will be asked to predict your Leader's decisions in Stage 1. Specifically, we would like to know how likely it is in your opinion that the Leader has chosen Investment X in each of the five investment tasks in Stage 1.

For each investment task, the specific questions you will be asked are listed below.

> Question 1
> **How likely do you think it is that your Leader has chosen Investment X? Specifically, what is the chance out of 100 that s/he has chosen Investment X?**

In Question 2, you are given additional information. You are asked to evaluate the same question with this additional information.

> Question 2(a)
> **Suppose you are informed that the investment chosen by your Leader has succeeded. This gives you a high payoff.**
>
> Now consider whether your prediction will be higher than, lower than, or the same as the one you stated in Question 1.
>
> Specifically, given that the investment has succeeded, what is the chance out of 100 that s/he has chosen Investment X?
>
> Question 2(b)
> **Suppose you are informed that the investment chosen by your Leader has failed. This gives you a low payoff.**
>
> Now consider whether your prediction will be higher than, lower than, or the same as the one you stated in Question 1.
>
> Specifically, given that the investment has failed, what is the chance out of 100 that s/he has chosen Investment X?

For both questions, you will need to choose a number between 0 and 100. A higher number means that you think your Leader is more likely to have chosen Investment X.

For your payment, the computer will randomly select one of these two questions and you will be paid for your response to this question. If Question 2 is chosen for payment, then you will be paid for your answer to the scenario that corresponds to the actual outcome, i.e., you will be paid for Question 2(a) if the investment has succeeded or Question 2(b) if it has failed.

To determine your payment, we use a procedure which has been used in many other studies. We explain the procedure in detail, but what is most important is that this payoff structure is designed such that it is in your best interest to report your true belief about the chance that your Leader has chosen Investment X.

Your payment will be determined as follows. You will receive either 0 ECU or 200 ECU. Your chance of receiving 200 ECU depends on your prediction and the actual decision made by your Leader.

Specifically, your chance of receiving 200 ECU is determined by the following formulas:

Chance of receiving 200 ECU if Leader chose **Investment X**

$$= \left[1 - \left(\frac{100-\text{prediction}}{100}\right)^2\right] \times 100$$

Chance of receiving 200 ECU if Leader chose **Investment Y**

$$= \left[1 - \left(\frac{\text{prediction}}{100}\right)^2\right] \times 100$$

Suppose you state a high number as your prediction that your Leader chose Investment X. The formulas above imply that your chance of receiving 200 ECU is high if s/he chose Investment X, and your chance of receiving 200 ECU is low if s/he chose Investment Y. Hence, you should carefully consider how likely it is that your Leader chose Investment X.

To illustrate, suppose your prediction that your Leader chose Investment X is 100. Then, if s/he chose Investment X, your chance of receiving 200 ECU will be $\left[1 - \left(\frac{100-100}{100}\right)^2\right] \times 100 = 100$. If s/he chose Investment Y, your chance of receiving 200 ECU will be $\left[1 - \left(\frac{100}{100}\right)^2\right] \times 0$. Hence, your prediction should depend on whether you think your Leader is more likely to have chosen Investment X or Investment Y.

Here are two more examples explaining how your chance of receiving 200 ECU will be determined based on your prediction and the decision made by your Leader.

Example 1: Suppose you predict 70 as the chance that your Leader chose Investment X. At the end of the experiment, the computer reveals that s/he chose Investment X. Then, your chance of receiving 200 ECU will be $\left[1-\left(\frac{100-70}{100}\right)^2\right] \times 100 = 91$.

Example 2: In the above example, suppose your Leader chose Investment Y. Then, your chance of receiving 200 ECU will be $\left[1-\left(\frac{70}{100}\right)^2\right] \times 100 = 51$.

To determine whether you receive 200 ECU, the computer will randomly draw a number between 0 and 100. Each number between 0 and 100 is equally likely to be picked. If the number drawn by the computer is less than or equal to your chance of receiving 200 ECU as determined by the formulas above, then you will receive 200 ECU. Otherwise, you will receive 0 ECU. Hence, in Example 1 above, if the number randomly drawn by the computer is less than or equal to 91, then you will receive 200 ECU. Otherwise, you will receive 0 ECU.

In summary, your prediction will determine the chance that you receive 200 ECU. The closer your prediction is to the actual decision of your Leader, the higher your chance is of receiving 200 ECU.

Decisions to modify the Leader's payoff: After you state your predictions, you will be asked whether you would like to modify your Leader's payoff from Stage 1, given each possible outcome of the investment chosen by him/her.

Specifically, you may choose to either increase or decrease your Leader's payoff by an amount between 10 ECU and 100 ECU. You may choose any amount in multiples of 10 ECU within this range. You may also choose not to increase or decrease your Leader's payoff, i.e., you may choose to modify your Leader's payoff by 0 ECU.

You will be asked to make this decision both assuming that the investment chosen by your Leader has succeeded, and assuming that the investment chosen by your Leader has failed. <u>Note that your earnings will not be affected by your decisions to increase or decrease your Leader's payoff.</u>

At the end of the experiment, the decisions of one of the two Members within the group will be randomly selected to be implemented. Your Leader's payoff from Stage 1 will then be modified according to the decision that corresponds to the actual outcome of the investment.

Here is a scenario and two examples to illustrate how your decisions as a Member will affect your Leader's payoff from Stage 1.

Scenario: Suppose you choose to increase your Leader's payoff by 60 ECU if the investment succeeds, and decrease it by 80 ECU if the investment fails. The other Member of your group chooses to modify your Leader's payoff by 0 ECU if the investment succeeds, but increase it by 30 ECU if the investment fails.

Example 3: At the end of the experiment, suppose the computer reveals that the investment chosen by your Leader has failed, and that your decisions have been implemented. In this case, your Leader's payoff from Stage 1 will be decreased by 80 ECU.

Example 4: At the end of the experiment, suppose the computer reveals that the investment chosen by your Leader has succeeded, and that the decisions of the other Member have been implemented. In this case, your Leader's payoff from Stage 1 will not be modified.

Payment for Part 1

At the end of the experiment, the computer will randomly determine **one** of the five investment tasks for payment. For that randomly chosen investment task:

1. If you are the **Leader**, you will be paid according to your investment decision and the cost of that decision in Stage 1. Your payoff from Stage 1 may be modified based on your Members' decisions in Stage 2.

2. If you are a **Member**, the computer will randomly determine whether you will be paid for your Leader's investment decision in Stage 1 or your prediction of his/her decision in Stage 2.

Table 1 below summarizes the payoffs of the Leader and Members for each investment task.

Table 1: Payoffs to Leader and Members for each investment task of Part 1

	Paid for:	
	Investment Return	**Prediction**
Leader	Yes	Not applicable
Member	Either one but not both	

Summary

1. In Part 1, you will be divided into groups of three. There are two stages in Part 1. In each stage, you will be asked to make decisions relating to five investment tasks.

2. In Stage 1, you will be asked to make a decision for each investment task assuming that you are the Leader. As a Leader, you will be given an endowment of 300 ECU for each task and asked to choose between two investment options. Your choice will affect both your payoff and the payoffs of the Members you have been matched with. Your decisions in Stage 1 will be implemented only if you are assigned to be the Leader of your group.

3. The returns of Investment X and Investment Y may be different for each task. However, the investments always provide a higher return if they succeed and a lower return if they fail.

4. At the end of Stage 1, you will be provided information about your groups and roles. One participant in the group will be the Leader and the other two participants will be Members. You will be informed whether you are the Leader or a Member of your group after you have completed Stage 1 and before Stage 2 begins.

5. In Stage 2, if you are a Member, you will be asked to predict your Leader's decisions for the five investment tasks in Stage 1. For each investment task, you will be asked two questions.

   In Question 1, you will be asked to predict how likely it is in your opinion that your Leader has chosen Investment X. You will need to choose a number between 0 and 100. A higher number means that you think that s/he is more likely to have chosen Investment X.

   In Question 2, you will be asked the same question under two different scenarios: (i) assuming that the investment has succeeded; and (ii) assuming that the investment has failed. You should consider whether your prediction of your Leader's decision will be higher than, lower than, or the same as the one you stated in Question 1, given that you know the outcome of the investment chosen by him/her.

6. As a Member, the payoff structure used to determine your payment for your pre-dictions is designed such that it is in your best interest to report your true belief about the chance that your Leader has chosen Investment X.

7. After you state your predictions, you will be asked whether you would like to modify your Leader's payoff from Stage 1, given each possible outcome of the investment chosen by him/her.

   You may choose to either increase or decrease your Leader's payoff by an amount between 10 ECU and 100 ECU. You may choose any amount in multiples of 10 ECU within this

range. You may also choose not to modify your Leader's payoff. Your earnings as a Member will not be affected by your decisions to modify your Leader's payoff.

8. At the end of the experiment, the computer will randomly select one of the five investment tasks for payment. For the randomly chosen investment task:

   (a) The Leader will be paid for their investment decision in Stage 1, and their payoff may be modified based on the Members' decisions in Stage 2.

   (b) Each Member will be paid either for their Leader's decision in Stage 1 or their prediction of the Leader's decision in Stage 2.

If you have any questions, please raise your hand and an experimenter will come to you to answer your questions privately. Otherwise, please proceed to answer the practice questions on your computer screen. The purpose of these practice questions is to make sure that you understand the experiment.

When you are ready to begin the practice questions, please press the button on your computer screen to launch the practice questions.

Part 2

You will participate in Part 2 in groups of two. The computer will randomly match you with one other person in the room. You will never learn the identity of your partner.

Each of you is given an endowment of 300 ECU, and you are asked to divide this amount between yourself and the person you are matched with.

At the end of today's session, if Part 2 is picked for payment, then you will be paid either according to your decision or according to the decision made by your randomly matched partner. The computer will randomly determine whose allocation decision will be implemented.

Example. Suppose you choose to divide your endowment by keeping 200 ECU for yourself and giving 100 ECU to your matched partner. Your matched partner decides to keep 130 ECU and give 170 ECU to you. If, at the end of the experiment, the computer randomly determines that it is the allocation of your matched partner that gets implemented, then your payment will be 170 ECU and your matched partner's payment will be 130 ECU.

Are there any questions? If not, we will proceed with Part 2.

Part 1: Practice Questions

(These are programmed on z-Tree.)

1. Which of the following statements is correct?
   (a) I will be paid for the decisions in both parts of the experiment today.
   (b) I will be paid for the decisions in either Part 1 or Part 2 of the experiment today.

   Answer: (b)

2. We will make decisions relating to 5 investment tasks in Part 1. If we are paid for Part 1, then we will be paid for our decisions for one of the 5 investment tasks.
   (a) True
   (b) False

   Answer: (a)

3. In Stage 1 of Part 1, everyone will make decisions as Leaders.
   (a) True
   (b) False

   Answer: (a)

4. In Stage 2 of Part 1,
   (i) I will learn whether I am the Leader or a Member of my group.
   (ii) everyone will make decisions as Members.

   (a) Both (i) and (ii) are correct.
   (b) (i) is correct but (ii) is incorrect.
   (c) (i) is incorrect but (ii) is correct.
   (d) Both (i) and (ii) are incorrect.

   Answer: (b)

5. Which of the following statements is correct?
   (a) The Members will be informed of the investment chosen by the Leader, but not the outcome of the investment.
   (b) The Members will be informed of the outcome of the investment chosen by the Leader, but not the investment chosen by him/her.
   (c) The Members will be informed of the investment chosen by the Leader, and the outcome of the investment.

   Answer: (b)

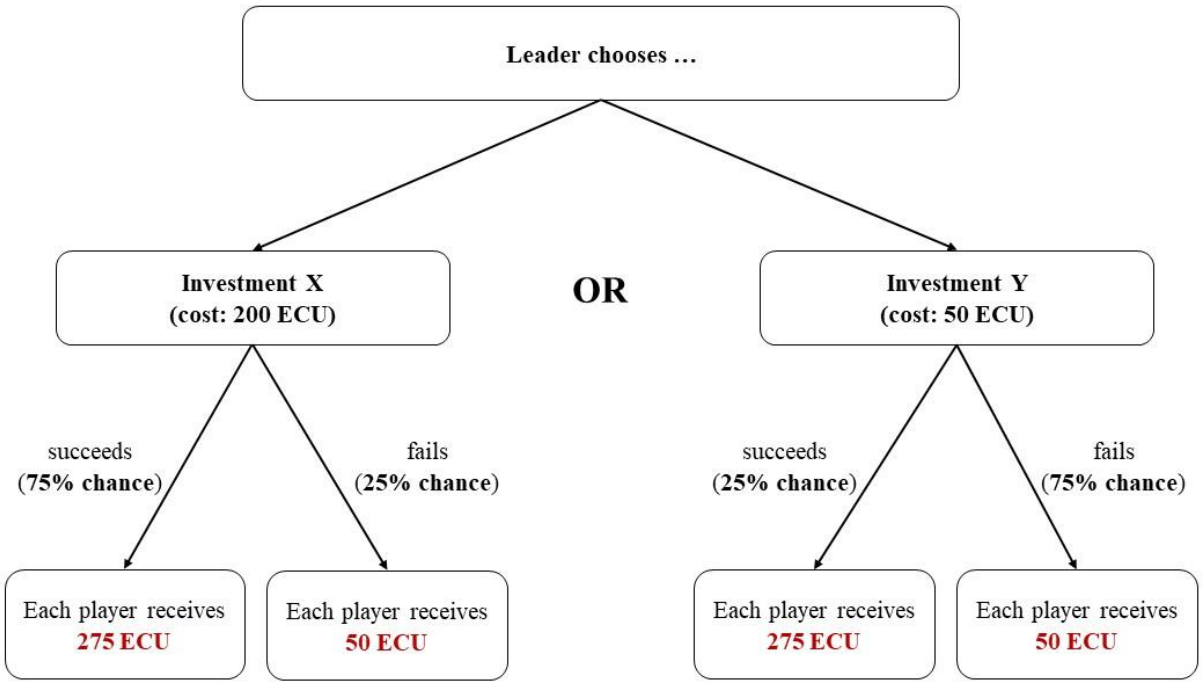
6. If I am a Member, then I will be paid for:
   (a) my Leader's investment decision in Stage 1 only.
   (b) my prediction of my Leader's investment decision in Stage 2 only.
   (c) both my Leader's investment decision in Stage 1 AND my prediction of his/her decision in Stage 2.
   (d) either my Leader's investment decision in Stage 1 OR my prediction of his/her decision in Stage 2, but not both.

   Answer: (d)

7. If I am a Member, I will be asked two questions. If I am paid for my predictions, then I will be paid accordingly to my responses to both questions.
   (a) This statement is true.
   (b) This statement is false. I will be asked only one question as a Member. If I am paid for my predictions, then I will be paid for my response to that question.
   (c) This statement is false. I will be asked two questions as a Member. However, if I am paid for my predictions, then I will be paid for my response to only one of the questions.

   Answer: (c)

Consider the investment options below.



8. Suppose the Leader chooses **Investment X**.

   At the end of the experiment, the computer randomly picks this task for payment and determines that the investment fails. What are the net payoffs of the Leader and each Member in Stage 1?

   Answer:
   Leader: 150
   Each Member: 50

9. Suppose the Leader chooses **Investment Y**.

   At the end of the experiment, the computer randomly picks this task for payment and determines that the investment succeeds. What are the net payoffs of the Leader and each Member in Stage 1?

   Answer:
   Leader: 525
   Each Member: 275

10. Suppose you are a Member. If you strongly believe that your Leader has chosen Investment Y, which of the following statements is true?

    (a) It is in my best interest to choose a high number as my prediction of the chance that my Leader has chosen Investment X.
    (b) It is in my best interest to choose a low number as my prediction of the chance that my Leader has chosen Investment X.
    (c) It is in my best interest to choose 50 as my prediction of the chance that my Leader has chosen Investment X.

    Answer: (b)

11. If I am a Member, then in Stage 2,
    (i) my decisions to increase or decrease my Leader's payoff from Stage 1 will affect my own earnings.
    (ii) I can choose not to modify my Leader's payoff from Stage 1.

    (a) Both (i) and (ii) are correct.
    (b) (i) is correct but (ii) is incorrect.
    (c) (i) is incorrect but (ii) is correct.
    (d) Both (i) and (ii) are incorrect.

    Answer: (c)

Experimenter Notes

Before we proceed to Stage 2, we will now announce your groups and roles. Please pay attention to your computer screens.

(LAUNCH NEXT SCREEN)

You can see on your screen the ID number assigned to you prior to this experiment.

Remember that you have been divided into groups of three. One participant in the group is the Leader. The other two participants are Members. In a few moments, you will be informed on your computer screens your group number, and whether you are the Leader or a Member.

To ensure that all participants have been assigned to a group of three, we will announce each group separately. When we call out your group number, please raise your hand (above the partition) so that I can see it.

To ensure that every group has a Leader, we will also announce the Leader in each group by calling out the last three digits of their ID number. If you are the Leader, when I call out the last three digits of your ID number, please loudly and clearly announce "Here". Please only say "Here" and nothing else.

To maintain your anonymity, please remain seated and face your computer screens.

Does anyone have any questions? If not, we will now begin with Group 1.

(LAUNCH NEXT SCREEN)

If you are in Group X, you will see this information on your screens. Please raise your hand if you are in Group X.

Please put down your hands.

(LAUNCH NEXT SCREEN)

I will now announce the Leader. The Leader in Group X has an ID number ending in: XXX.

(AFTER ALL GROUPS REVEALED)

We will now proceed with Stage 2.

Appendix B Conceptual Framework

We present a simple model illustrating how leaders and evaluators are expected to make decisions in the context of our experiment.

B1. Leaders

In line with our experimental design, we consider an environment where each leader makes a discrete investment choice of X or Y on behalf of a group of N players. The leader's investment choice is not observable and results in an outcome for the group which we represent as output Q, where $Q \in \{Q_L, Q_H\}$ and $Q_H > Q_L$. The realized output level is equally shared between the group members (including the leader) although the investment cost is solely borne by the leader. Output is probabilistic and the leader's investment choice affects the probability of a high-output realization (Q_H). Specifically, assume that an investment choice of X leads to Q_H with probability $p \in (0.5,1)$ and a choice of Y leads to Q_H with probability $(1-p)$. Hence, investment choice X (which is more costly for the leader) leads to a high output level with a higher probability. The investment cost $c \in \{c_X, c_Y\}$ is deducted from an initial endowment $\omega \geq c_X > c_Y$ that the leader receives.

Leaders are differentiated based on the weight they put on their own monetary payoff (λ_i) and the weight they put on others' (i.e., their group members') payoff ($\alpha_i < \lambda_i$). $\alpha_i = 0$ stands for a purely self-interested leader. Both λ_i and α_i are private draws from a joint distribution $F(\lambda, \alpha)$ which is common knowledge. For a given outcome realization and discretionary payment amount, we write leader i's utility as:

$$\lambda_i u_i\left(\frac{Q}{N} + \omega - c + \Delta_j\right) + \alpha_i \sum_j v_j\left(\frac{Q}{N}\right), \tag{B.1}$$

where u_i and v_j are twice differentiable utility functions. We assume that u_i represents the direct utility leader i receives from his/her own monetary payoff and v_j is the utility evaluator j receives from his/her own monetary payoff. Δ_j stands for the discretionary payment that evaluator j pays to leader i. Each leader makes the investment choice with the objective of maximizing his/her expected utility.

B2. Evaluators

In our experimental design, the group members decide whether they would like to make a discretionary payment to the leader after observing the outcome (Q) but not the investment choice. Although each leader makes decisions for five investment tasks (with different parameterizations) and each evaluator is asked to make five discretionary payment decisions, leaders do not receive any information about the decisions of the evaluators during the

experiment. This implies discretionary payments cannot be motivated by an incentive to change the future behavior of leaders, so we model them as reciprocal actions.

We consider a model of reciprocity where reciprocal actions are potentially determined by both (perceived) intentions and outcomes.[1] Making this distinction and considering both factors are important given the evidence from the literature which shows that reciprocal behavior can be shaped by both intentions (see, e.g., Charness, 2004; Charness and Levine, 2007) and outcomes (e.g., Gurdal, Miller, and Rustichini, 2013; Brownback and Kuhn, 2019).

Specifically, to determine the evaluators' reciprocal actions, we introduce a kindness term as a function of perceived intentions and outcomes. Let φ_i denote evaluator j's perception of leader i's kindness, and $\rho_j > 0$ represent the reciprocity preference of evaluator j. Each evaluator chooses Δ_j to maximize his/her utility function given his/her reciprocity preference (ρ_j) and his/her perception of the leader's kindness (φ_i).

We refer to the evaluator's beliefs about the leader's intentions as "perceived intentions." In the experiment, to capture these beliefs, we ask the evaluators their beliefs on the likelihood that the leader has chosen Investment X. As we show below, this is equivalent to eliciting the evaluators' beliefs about their leader's type. Evaluators update their prior beliefs after observing the output. Let $\sigma_j(\cdot|Q)$ stand for the updated (posterior) belief that evaluator j has about the leader's type after observing the output Q.

We assume that the kindness term takes the following form: $\varphi_i = \theta_j\left(1_{Q=Q_H} - 1_{\ Q=Q_L}\right) + \beta_j\left(2\sigma_j(\cdot|Q) - 1\right)$, where $\theta_j \in [0,1]$ and $\beta_j \in [0,1]$ are the weights evaluator j puts on outcomes and perceived intentions, respectively, when determining the leader's kindness, and $1_{Q=Q_H}$ and $1_{Q=Q_L}$ are indicator functions for whether a high or a low output is observed. This specification implies that $\left(1_{Q=Q_H} - 1_{\ Q=Q_L}\right) \in \{-1,1\}$, $\left(2\sigma_j(\cdot|Q) - 1\right) \in [-1,1]$, and, consequently, $\varphi_i \in [-2,2]$. Evaluator j views leader i as kind if $\varphi_i > 0$ and as unkind if $\varphi_i < 0$.

The kindness term indicates that if two evaluators care about intentions only (i.e., $\theta_j = 0$ and $\beta_j > 0$) and if they have the same posterior beliefs, then they will choose the same discretionary payment irrespective of the outcome. On the other hand, if two evaluators care

[1] See Falk and Fischbacher (2006). Unlike Falk and Fischbacher (2006), we have a model where agents have private information about their types. The reciprocator's perception of the agent's underlying intention is determined by the reciprocator's belief about the agent's type.

about outcomes only (i.e., $\beta_j = 0$ and $\theta_j > 0$) and if they observe different outcomes, then they will choose different discretionary payments even if their posterior beliefs are the same. In this case, their decisions will purely be guided by outcome bias. Hence, in this framework, outputs potentially affect discretionary payments through two different channels: in addition to the indirect impact they have through evaluators' posterior beliefs, they can also have a direct impact independent of the beliefs.

Each evaluator has private information about his/her three-dimensional type represented by ρ_j, θ_j and β_j. Let $G(\rho, \theta, \beta)$ stand for the joint distribution function from which types are drawn. $G(\rho, \theta, \beta)$ is common knowledge. After the output is realized, we assume evaluator j chooses Δ_j to maximize the following utility function:

$$u_j\left(\frac{Q}{N}\right) + \rho_j \varphi_i \Delta_j - c(\Delta_j), \tag{B.4}$$

where $\varphi_i = \theta_j\left(1_{Q=Q_H} - 1_{Q=Q_L}\right) + \beta_j\left(2\sigma_j(\cdot|Q) - 1\right)$ as defined above, and $c(\Delta_j)$ stands for the cost of making a discretionary payment decision. The cost of making a discretionary payment decision may be monetary or non-monetary (e.g., psychological). We assume that $c(\Delta_j)$ is increasing and convex in Δ_j.

B3. Equilibrium

We use Perfect Bayesian Equilibrium as the equilibrium concept. To characterize the equilibrium, we need to specify:

(i) $i(\cdot): [0,1] \to \{X, Y\}$, i.e., the leader's investment as a function of the leader's type;

(ii) $\sigma_j(\cdot|Q)$, i.e., the probability density function summarizing the posterior beliefs evaluators have after observing the output Q;

(iii) $\Delta(\cdot,\cdot): \mathbb{R}_+^3 \text{ x } \{Q_L, Q_H\} \to \mathbb{R}$, i.e., the discretionary payment which is a mapping from the evaluator's (three dimensional) type and outcome.

The leader makes the investment choice by comparing the expected utility from choosing Investment X with the expected utility from choosing investment Y. The expected utility from choosing X is given by:

$$p\left(\lambda_i u_i\left(\frac{Q_H}{N} + \omega - c_H + E_{\rho,\theta,\beta}(\Delta_j|Q_H)\right) + \alpha_i \sum_j v_j\left(\frac{Q_H}{N}\right)\right)$$
$$+ (1-p)\left(\lambda_i u_i\left(\frac{Q_L}{N} + \omega - c_L + E_{\rho,\theta,\beta}(\Delta_j|Q_L)\right) + \alpha_i \sum_j v_j\left(\frac{Q_L}{N}\right)\right)$$

where $E_{\rho,\theta,\beta}(\Delta_j|Q_H)$ and $E_{\rho,\theta,\beta}(\Delta_j|Q_L)$ stand for the discretionary payments that the leader expects to receive when the output realizations are high and low, respectively. The expected

utility from choosing investment Y can be written in a similar way. It is straightforward to show that the leader will choose Investment X over Investment Y if the following condition holds:

$$p\left(\lambda_i u_i^H + \alpha_i \sum_j v_j^H\right) + (1-p)\left(\lambda_i u_i^L + \alpha_i \sum_j v_j^L\right)$$
$$\geq (1-p)\left(\lambda_i u_i^H + \alpha_i \sum_j v_j^H\right) + p\left(\lambda_i u_i^L + \alpha_i \sum_j v_j^L\right)$$

where u_i^H and u_i^L stand for the leader's utility from his/her own monetary payoff when the output realization is high and low, respectively. Similarly, v_j^H and v_j^L are defined as the utility evaluator j receives when the output realization is high and low, respectively. Simplifying yields

$$\frac{\alpha_i}{\lambda_i} \geq \frac{(u_i^L - u_i^H)}{\left(\sum_j v_j^H - \sum_j v_j^L\right)}$$

This condition shows that if the expected discretionary payments are sufficiently high such that $u_i^L < u_i^H$, then all leader types will choose Investment X. However, if the expected discretionary payments are such that $u_i^L > u_i^H$, then there will be a leader type who is indifferent between Investment X and Investment Y.

Let $\kappa_i \equiv \frac{\alpha_i}{\lambda_i}$ stand for the relative importance leaders give to others' payoff (their relative prosociality) and let κ^* stand for the indifferent leader type. Then, all leaders with $\frac{\alpha_i}{\lambda_i} > \kappa^*$ would prefer Investment X. In other words, the optimal investment choice partitions the interval [0,1] into two sets:

$$T_L = \{\kappa \in [0,1]: i = \text{X}\} \text{ and } T_H = \{\kappa \in [0,1]: i = \text{Y}\}$$

Type κ^* stands at the intersection of T_L and T_H such that $T_L = [0, \kappa^*]$ and $T_H = [\kappa^*, 1]$.

To form beliefs about the choices made by the leaders, evaluators form second-order beliefs on the leaders' expectations of discretionary payments.[2] Given these beliefs, they can solve for the leader type who is indifferent between choosing Investment X and Investment Y. Since leader i will choose Investment X in equilibrium if his or her type is above the threshold value κ^*, the prior belief an evaluator has about the leader choosing Investment X is $1 - H(\kappa^*)$,

[2] In a rational expectations equilibrium, evaluators would expect the leaders' expectations to be equal to actual payments. However, in a behavioral model, evaluators may believe leaders' expectations to be shaped by social norms.

where H represents the distribution of κ. Assuming Bayesian updating, the probability density functions for the posterior beliefs (after observing the output levels) are given by:

$$\sigma(\kappa|Q_H) = \begin{cases} \dfrac{(1-p)h(\kappa)}{\big(1-H(\kappa^*)\big)p + H(\kappa^*)(1-p)} & \text{for } \kappa \in T_L \\ \dfrac{ph(\kappa)}{\big(1-H(\kappa^*)\big)p + H(\kappa^*)(1-p)} & \text{for } \kappa \in T_H \end{cases}$$

$$\sigma(\kappa|Q_L) = \begin{cases} \dfrac{ph(\kappa)}{\big(1-H(\kappa^*)\big)(1-p) + H(\kappa^*)p} & \text{for } \kappa \in T_L \\ \dfrac{(1-p)h(\kappa)}{\big(1-H(\kappa^*)\big)(1-p) + H(\kappa^*)p} & \text{for } \kappa \in T_H \end{cases}$$

Finally, evaluators determine their discretionary payments which are given by

$$\Delta_j \in \arg\max \rho_j \Delta_j \big[\theta_j \big(1_{Q=Q_H} - 1_{Q=Q_L}\big) + \beta_j \big(2\sigma_j(\kappa|Q) - 1\big)\big] - c\big(\Delta_j\big)$$

Evaluators maximize their utility by responding to kind actions ($\varphi_i > 0$) with a positive discretionary payment that increases the payoff of the leader and to unkind actions ($\varphi_i < 0$) with a negative discretionary payment that decreases the payoff of the leader.

B4. Leaders' Gender

Male and female leaders may differ both in terms of the distributions from which they draw their types and in terms of their expectations regarding discretionary payments. If male and female leaders have different expectations in terms of discretionary payments, then the indifferent type (between Investment X and Y) for male leaders may be different from the indifferent type for female leaders. The evaluators' beliefs on the likelihood that the leader has chosen Investment X will depend on these thresholds as well as the underlying type distributions.

Gender differences in discretionary payments may stem from gender differences in beliefs or perceived intentions represented by $\sigma(\kappa|Q)$, gender differences in the direct weight put on outcomes represented by θ_j, and gender differences in the weight put on perceived intentions represented by β_j. Evaluators use different criteria in the determination of the payments for male and female leaders if θ_j and β_j take on different values depending on the leader's gender. We refer to the gender difference we may observe in the criteria used as the *gender criteria gap*.

References

Brownback, A., Kuhn, M.A. (2019). Understanding outcome bias. *Games and Economic Behavior,* 117:342-360.

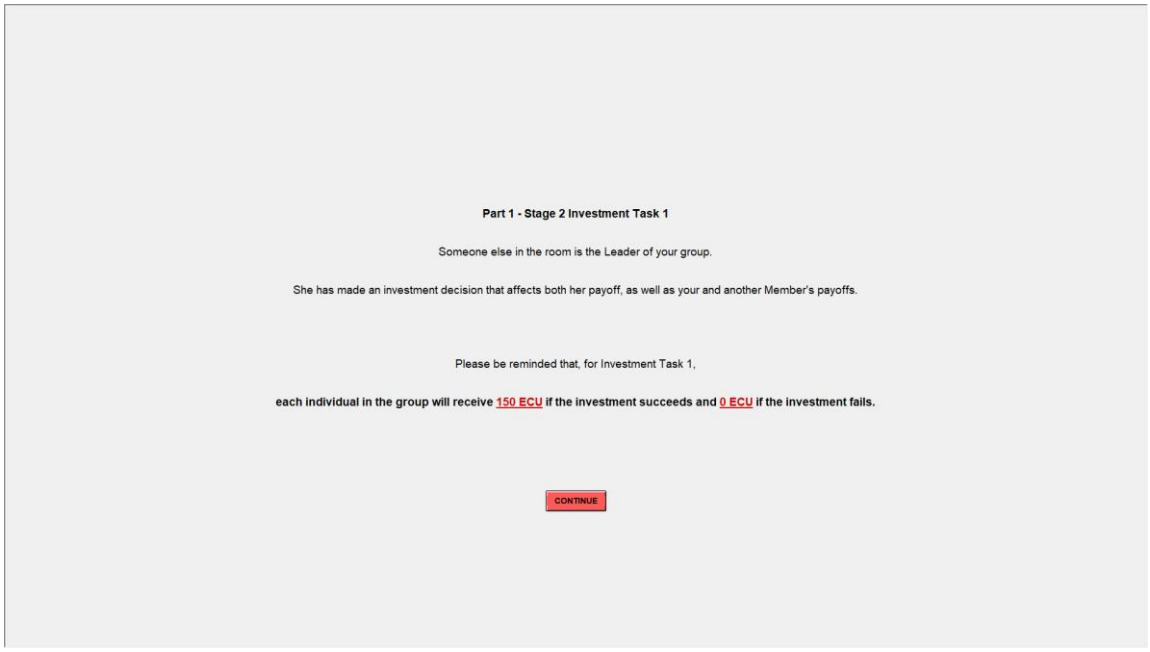
Charness, G. (2004). Attribution and reciprocity in an experimental labor market. *Journal of Labor Economics,* 22(3):665-688.

Charness, G., Levine, D.I. (2007). Intention and stochastic outcomes: An experimental study. *The Economic Journal,* 117(522):1051-1072.

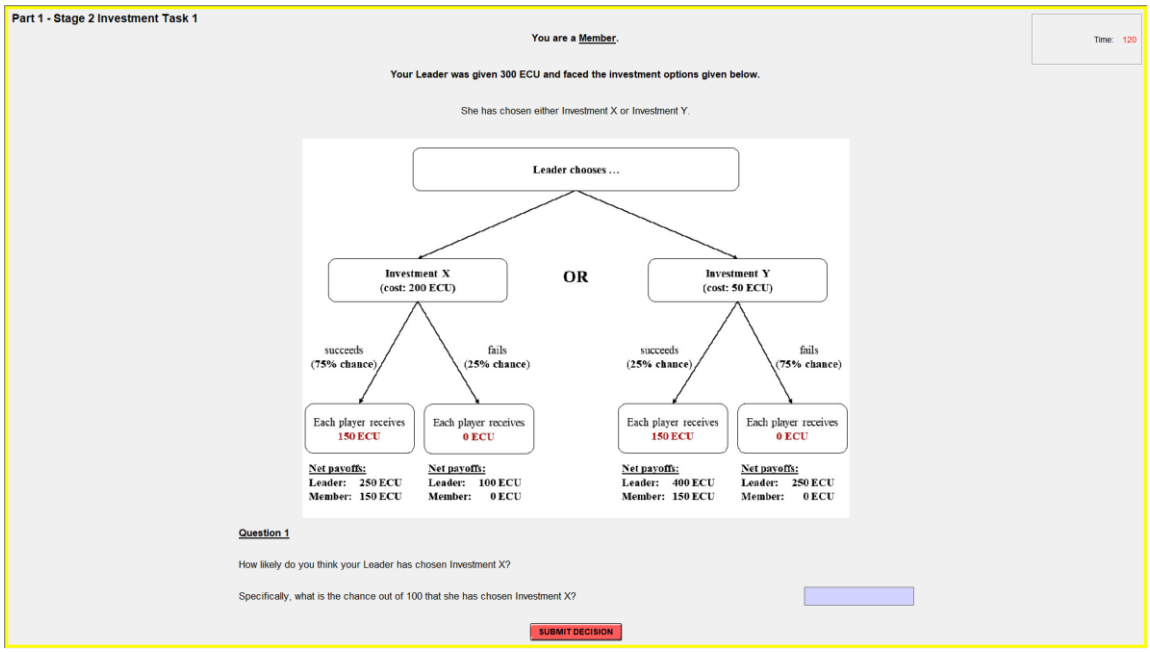
Falk, A., Fischbacher, U. (2006). A theory of reciprocity. *Games and Economic Behavior,* 54(2):293-315.

Gurdal, M.Y., Miller, J.B., Rustichini, A. (2013). Why blame? *Journal of Political Economy,* 121(6):1205-1247.

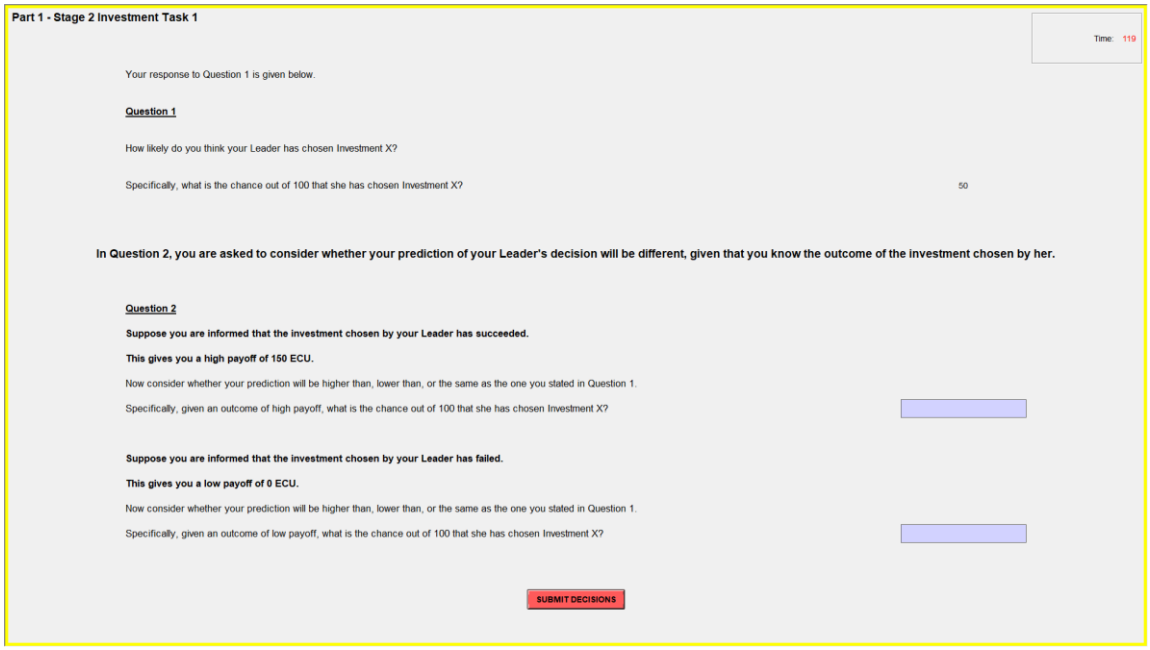
Appendix C Additional Tables and Figures



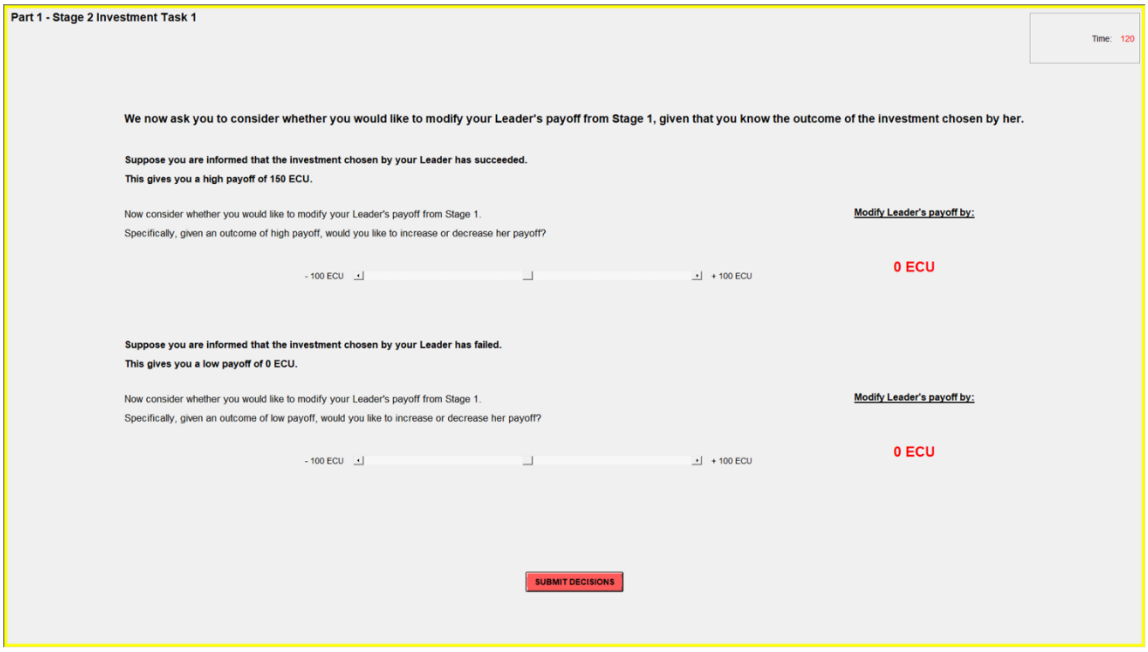
(a) Information about parameters of the investment task



(b) Prior beliefs elicitation screen



(c) Posterior beliefs elicitation screen



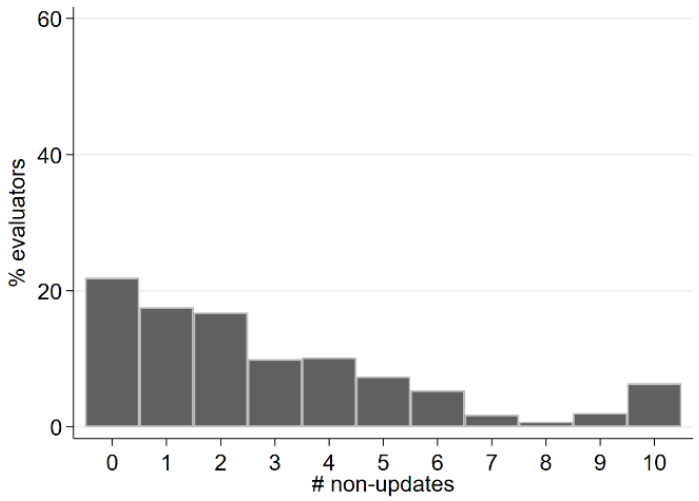
(d) Discretionary payment decisions screen

Figure C1: Example screenshots demonstrating the use of gender-specific pronouns to refer to the leader (female leader in this example)

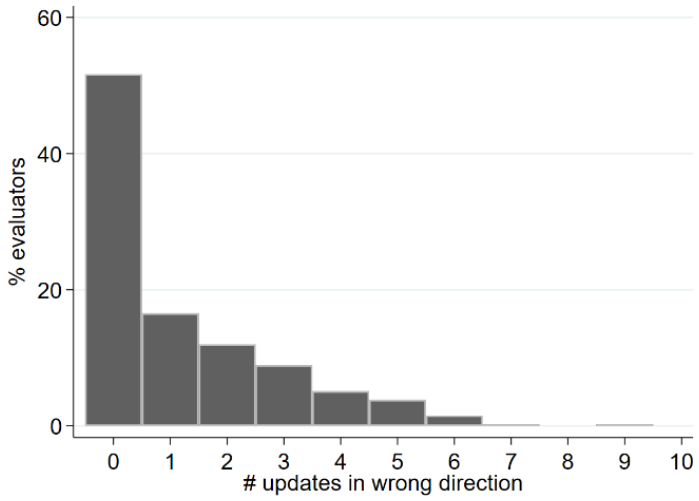
Table C1: OLS regressions of discretionary payments (pooling across both leader's gender)

	Dependent variable: Discretionary payments					
Variables	(1)	(2)	(3)	(4)	(5)	(6)
Female leader	7.418	9.080	5.180	7.238	11.633	13.657
	(7.608)	(7.497)	(7.738)	(7.628)	(8.707)	(8.504)
High outcome	40.465***	40.439***	40.894***	40.462***	35.438***	34.675***
	(4.841)	(4.862)	(5.265)	(5.306)	(5.702)	(5.730)
Female leader × High outcome	8.431	9.264	11.170	12.384*	15.296*	17.299**
	(6.467)	(6.528)	(6.861)	(6.939)	(8.279)	(8.483)
Posterior belief	0.367***	0.368***	0.367***	0.383***	0.507***	0.531***
	(0.095)	(0.095)	(0.100)	(0.101)	(0.103)	(0.103)
Female leader × Posterior belief	-0.228*	-0.261**	-0.257**	-0.302**	-0.354**	-0.416***
	(0.130)	(0.130)	(0.129)	(0.129)	(0.149)	(0.152)
Inconsistent or non-updater	-11.920**	-10.216**				
	(5.021)	(5.079)				
% endowment transferred in DG	0.410***	0.411***	0.322**	0.345**	0.295*	0.310*
	(0.141)	(0.140)	(0.144)	(0.143)	(0.164)	(0.161)
High Return - Low Return	-0.007	-0.006	-0.006	-0.006	-0.012	-0.012
	(0.010)	(0.010)	(0.011)	(0.011)	(0.013)	(0.013)
Zero return if investment fails	-1.516	-1.521	-1.309	-1.305	-0.735	-0.721
	(1.033)	(1.034)	(1.082)	(1.085)	(1.291)	(1.296)
Constant	-37.301***	-60.930***	-38.597***	-65.850***	-35.572***	-71.388***
	(9.501)	(15.256)	(10.274)	(16.038)	(11.071)	(18.039)
Control for task order	✓	✓	✓	✓	✓	✓
Control for Wave 2 data	✓	✓	✓	✓	✓	✓
Control for beliefs about leader's DG behavior	✓	✓	✓	✓	✓	✓
Individual controls		✓		✓		✓
Excluding non-updaters			✓	✓	✓	✓
Excluding inconsistent updaters					✓	✓
Observations	3,930	3,930	3,680	3,680	2,900	2,900
# participants (clusters)	393	393	368	368	290	290
R-squared	0.210	0.238	0.211	0.242	0.215	0.252

Robust standard errors clustered at the participant level in parentheses. Wave 2 data is the second wave of data collection conducted in 2025.
We classify an evaluator as a non-updater if all their posterior beliefs are equal to their prior beliefs for all five rounds of the investment task, and as an inconsistent updater if 25% or more of their posterior beliefs are in the opposite direction to that predicted by Bayes' rule. In columns (2), (4), and (6), we also control for participants' characteristics, which include their age, whether the participant is pursuing a major in economics, whether the participant is an undergraduate student, whether the participant is Australian, previous experience with economics experiments, and CRT score.
*** p<0.01, ** p<0.05, * p<0.10.



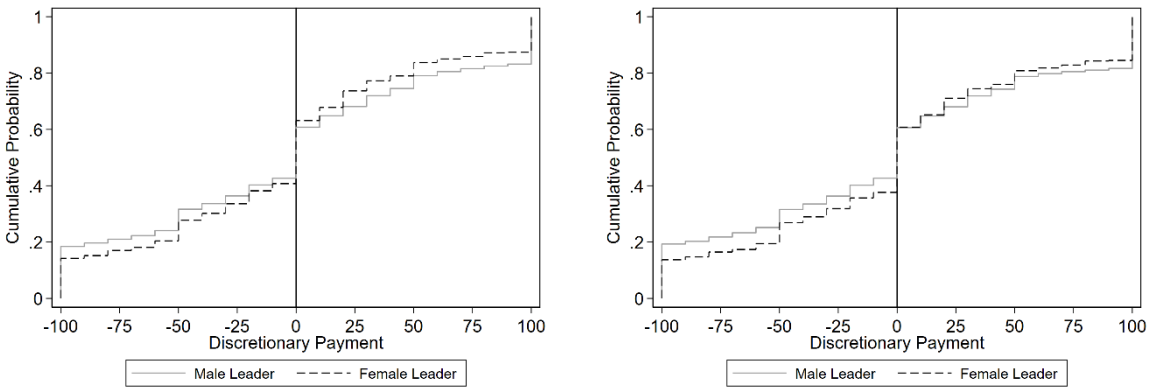
(a) Non-updates



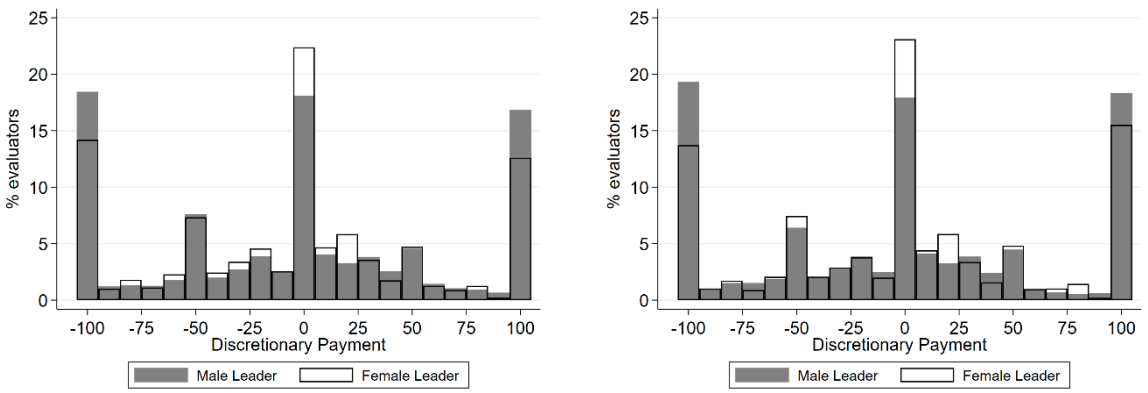
(b) Inconsistent updates

Figure C2: Distribution of non-updates and inconsistent updates by evaluators

Note: Panel (a) shows the distribution of non-updates made by evaluators. A non-update is defined as a posterior belief that is equal to the evaluator's prior belief. Panel (b) shows the distribution of inconsistent updates made by evaluators. An update is considered to be inconsistent if the evaluator's posterior belief moves in a direction opposite to that predicted by Bayes' rule. Note that there are five investment tasks, and evaluators report a posterior belief (update) conditional on a high outcome and a low outcome for each task. Hence, we observe 10 updates for each evaluator.



(a) Empirical cumulative distributions of discretionary payments
(left: excluding non-updaters; right: excluding non-updaters and inconsistent updaters)



(b) Histograms of discretionary payments
(left: excluding non-updaters; right: excluding non-updaters and inconsistent updaters)

Figure C3: Distributions of discretionary payments, by the leader's gender (excluding non-updaters and inconsistent updaters)

Note: Panel (a) presents the empirical cumulative distribution of discretionary payments for male leaders (solid line) and female leaders dashed line). The left-hand side graph excludes evaluators classified as non-updaters while the right-hand side graph excludes evaluators classified as non-updaters or inconsistent updaters. In both graphs, the distributions are statistically significantly different between genders (Kolmogorov-Smirnov test: p-value = 0.007 and 0.012, respectively). Panel (b) presents the corresponding distributions of payments as histograms and reveals that much of this difference is driven by female leaders being less likely than male leaders to receive the maximum penalty (-100 ECU) or the maximum bonus (+100 ECU).

Table C2: OLS regressions of evaluators' posterior belief that the leader has chosen Investment X, by both the leader's and evaluator's gender

	Dependent variable: Logit (posterior belief)					
	Female Evaluator			Male Evaluator		
Variables	Female Leader (1)	Male Leader (2)	(1) vs. (2) p-value	Female Leader (3)	Male Leader (4)	(3) vs. (4) p-value
(a) Full sample						
δ: Logit (prior belief)	0.642***	0.586***	0.606	0.520***	0.560***	0.754
	(0.072)	(0.080)		(0.092)	(0.090)	
γ_H: High outcome × logit (p)	0.627***	0.714*	0.656	0.929	0.965	0.871
	(0.112)	(0.160)		(0.162)	(0.148)	
γ_L: Low outcome × logit ($1-p$)	0.682**	0.878	0.362	0.718	0.769	0.842
	(0.140)	(0.164)		(0.184)	(0.177)	
$\gamma_H - \gamma_L$	-0.055	-0.165		0.211	0.196	
	(0.177)	(0.180)		(0.249)	(0.217)	
Observations	1010	980		980	950	
# participants (clusters)	101	98		98	95	
R-squared	0.485	0.472		0.360	0.377	
(b) Excluding non-updaters						
δ: Logit (prior belief)	0.513***	0.447***	0.503	0.437***	0.474***	0.764
	(0.061)	(0.078)		(0.092)	(0.083)	
γ_H: High outcome × logit (p)	0.656***	0.816	0.441	0.965	1.065	0.663
	(0.109)	(0.177)		(0.166)	(0.158)	
γ_L: Low outcome × logit ($1-p$)	0.750**	0.880	0.542	0.771	0.807	0.893
	(0.126)	(0.172)		(0.190)	(0.188)	
$\gamma_H - \gamma_L$	-0.095	-0.064		0.194	0.258	
	(0.138)	(0.203)		(0.253)	(0.233)	
Observations	940	920		930	880	
# participants (clusters)	94	92		93	88	
R-squared	0.361	0.340		0.297	0.312	
(c) Excluding non-updaters and inconsistent updaters						
δ: Logit (prior belief)	0.568***	0.457***	0.313	0.568***	0.599***	0.823
	(0.053)	(0.096)		(0.109)	(0.086)	
γ_H: High outcome × logit (p)	0.844	0.989	0.548	1.063	1.209	0.561
	(0.119)	(0.210)		(0.189)	(0.166)	
γ_L: Low outcome × logit ($1-p$)	0.993	1.101	0.637	1.084	1.002	0.764
	(0.130)	(0.188)		(0.189)	(0.201)	
$\gamma_H - \gamma_L$	-0.149	-0.112		-0.021	0.208	
	(0.154)	(0.234)		(0.235)	(0.244)	
Observations	740	740		700	710	
# participants (clusters)	74	74		70	71	
R-squared	0.468	0.378		0.470	0.425	

Robust standard errors clustered at the participant level in parentheses. Investment X refers to the costlier investment option for the leader but yields a higher success probability. We classify an evaluator as a non-updater if all their posterior beliefs are equal to their prior beliefs for all five rounds of the investment task, and as an inconsistent updater if 25% or more of their posterior beliefs are in the opposite direction to that predicted by Bayes' rule. Since the regression specification estimates parameters of an augmented Bayes' rule, no controls can be included as the presence of any controls would invalidate the interpretation of the parameters. Moreover, since $I(\text{High Outcome}) + I(\text{Low Outcome}) = 1$, there is no constant term in the regression.
*** p<0.01, ** p<0.05, * p<0.10. Null hypothesis is coefficient = 1.

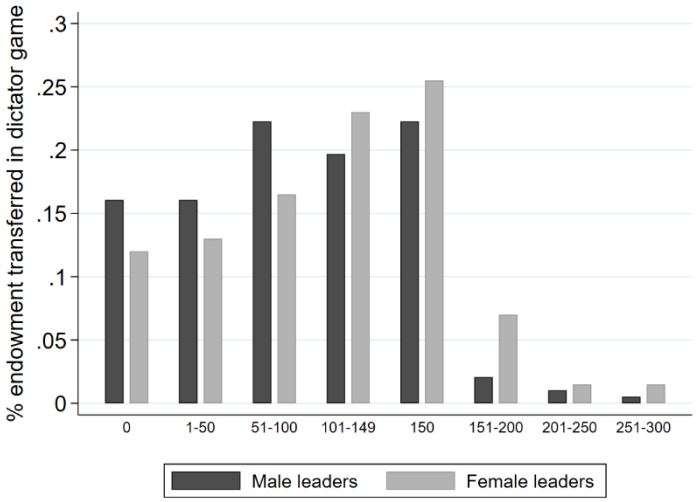
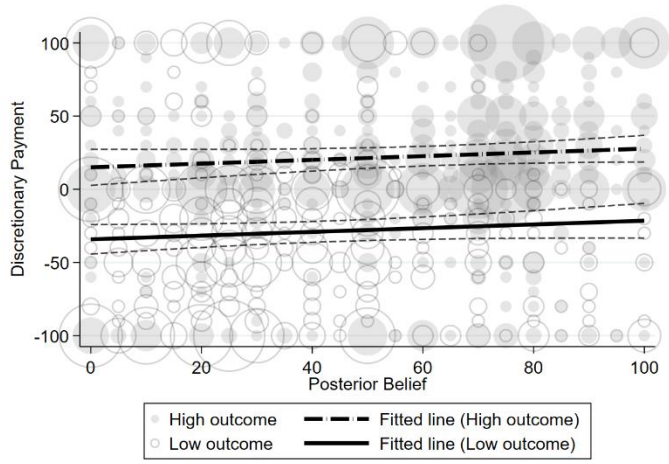
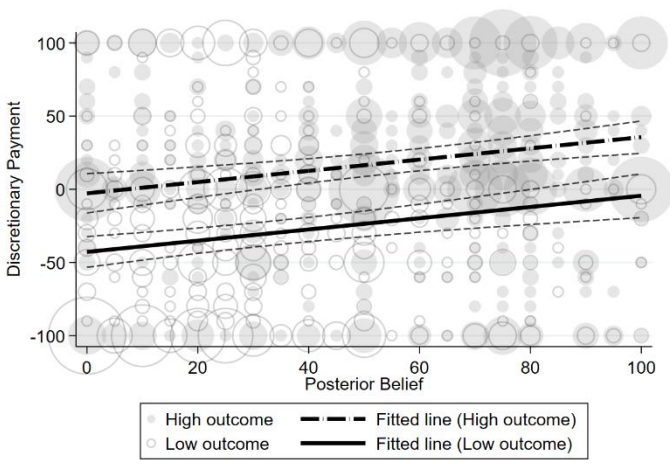


Figure C4: Evaluators' beliefs about amounts transferred by leaders in the dictator game, by the leader's gender

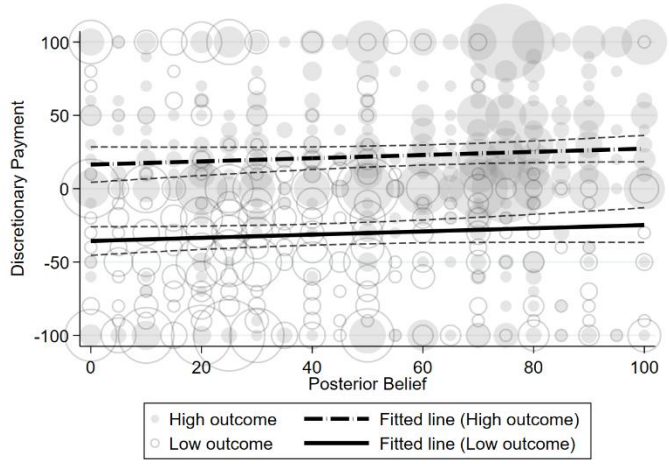
Note: The figure presents the distribution of evaluators' beliefs about the amount transferred by their leader in the dictator game, separately for male leaders (dark bars) and female leaders (light bars). Evaluators anticipate female leaders to transfer higher amounts to their matched partner in the dictator game than male leaders, with the different in distribution being marginally statistically significant (Kolmogorov-Smirnov test: p-value = 0.076).



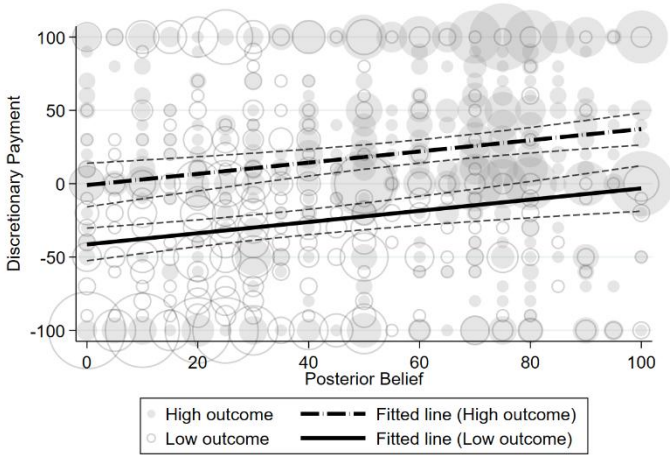
(a) Female Leaders (full sample)



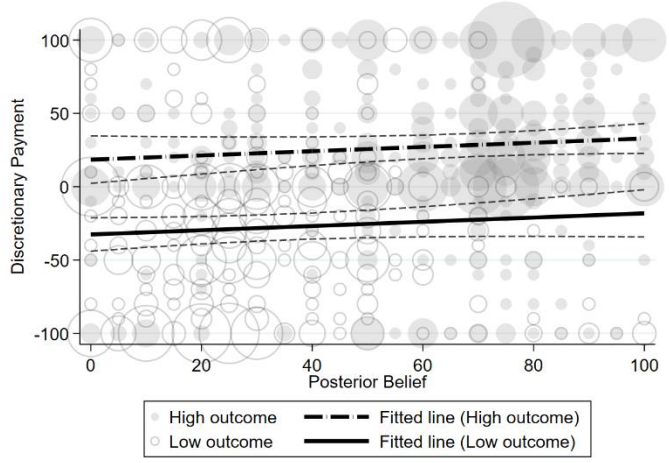
(b) Male Leaders (full sample)



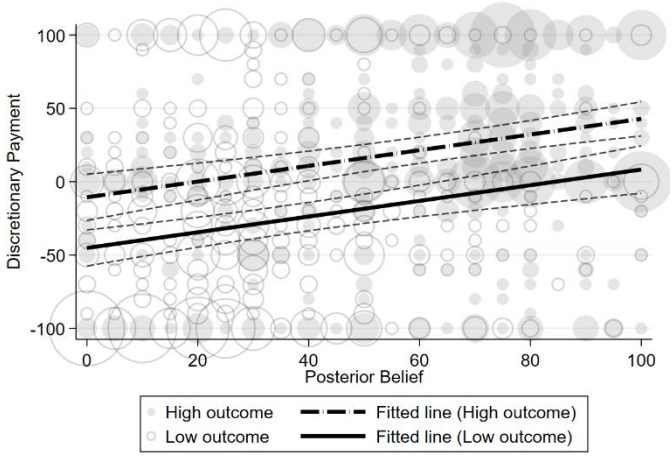
(c) Female Leaders (excluding non-updaters)



(d) Male Leaders (excluding non-updaters)



(e) Female Leaders
(excluding non-updaters and inconsistent updaters)



(f) Male Leaders
(excluding non-updaters and inconsistent updaters)

Figure C5: Bubble plots and fitted lines of discretionary payments against evaluators' posterior belief that the leader has chosen Investment X and leader's outcomes, by the leader's gender

Note: The figure presents fitted line plots of evaluators' discretionary payments against their posterior beliefs separately for female leaders (left 3 panels) and male leaders (right 3 panels). The lines are based on estimates of OLS regressions of evaluators' discretionary payments using equation (3), as presented in Table 4. Dashed lines above and below each fitted line represent 95% confidence intervals. The bubble plots of the discretionary payments are also included separately for high outcomes (gray bubbles) and low outcomes (white bubbles), where the size of each bubble is proportional to the number of observations.

Table C3: OLS regressions of discretionary payments, by the leader's gender (with controls for individual characteristics)

	Dependent variable: Discretionary payments								
Variables	Female Leader (1)	Male Leader (2)	p-value (1) vs. (2)	Female Leader (3)	Male Leader (4)	p-value (3) vs. (4)	Female Leader (5)	Male Leader (6)	p-value (5) vs. (6)
High outcome	49.923***	40.127***	0.128	52.941***	39.979***	0.060	52.187***	34.066***	0.028
	(4.193)	(4.942)		(4.364)	(5.411)		(5.952)	(5.817)	
Posterior belief	0.098	0.380***	0.029	0.076	0.400***	0.012	0.109	0.550***	0.003
	(0.085)	(0.099)		(0.079)	(0.103)		(0.105)	(0.105)	
Inconsistent or non-updater	-12.797*	-3.141							
	(6.597)	(7.703)							
% endowment transferred in DG	0.414**	0.475**		0.336*	0.383*		0.326	0.326	
	(0.175)	(0.214)		(0.187)	(0.219)		(0.218)	(0.234)	
High Return – Low Return	0.006	-0.020		0.012	-0.026		0.016	-0.041*	
	(0.013)	(0.017)		(0.013)	(0.018)		(0.015)	(0.021)	
Zero return if investment fails	-2.282*	-0.718		-2.075	-0.456		-1.432	0.016	
	(1.289)	(1.628)		(1.337)	(1.723)		(1.606)	(2.045)	
Constant	-62.434**	-54.268***		-75.995***	-48.246**		-72.076**	-56.848**	
	(25.102)	(18.391)		(25.110)	(20.030)		(29.215)	(23.136)	
Outcome/Belief: Coefficient	511.2	105.5		692.2	99.9		478.9	61.9	
Test of High outcome = 100 × Belief: p-value[(a)]	< 0.001	0.871		< 0.001	0.998		0.006	0.144	
Control for task order	✓	✓		✓	✓		✓	✓	
Control for Wave 2 data	✓	✓		✓	✓		✓	✓	
Control for beliefs about leader's DG behavior	✓	✓		✓	✓		✓	✓	
Individual controls	✓	✓		✓	✓		✓	✓	
Excluding non-updaters				✓	✓		✓	✓	
Excluding inconsistent updaters							✓	✓	
Observations	2,000	1,930		1,880	1,800		1,450	1,450	
# participants (clusters)	200	193		188	180		145	145	
R-squared	0.283	0.267		0.284	0.261		0.289	0.280	

Robust standard errors clustered at the participant level in parentheses. Wave 2 data is the second wave of data collection conducted in 2025.

We classify an evaluator as a non-updater if all their posterior beliefs are equal to their prior beliefs for all five rounds of the investment task, and as an inconsistent updater if 25% or more of their posterior beliefs are in the opposite direction to that predicted by Bayes' rule. In the regressions, we also control for participants' characteristics, which include their age, whether the participant is pursuing a major in economics, whether the participant is an undergraduate student, whether the participant is Australian, previous experience with economics experiments, and CRT score.

[(a)] This tests the null hypothesis that the coefficient on high outcome is equal to 100 times the coefficient on posterior beliefs. The interpretation of the test is whether there is a difference in the *marginal change* in evaluators' discretionary payments between two scenarios: (i) with respect to a given change in outcome (from a low outcome to a high outcome), and (ii) with respect to a given change in belief (from a belief that the leader has chosen Investment Y with certainty to a belief that the leader has chosen Investment X with certainty).

*** p<0.01, ** p<0.05, * p<0.10.

Table C4: OLS regressions of discretionary payments, by the leader's gender (against prior beliefs)

	Dependent variable: Discretionary payments								
Variables	Female Leader (1)	Male Leader (2)	p-value (1) vs. (2)	Female Leader (3)	Male Leader (4)	p-value (3) vs. (4)	Female Leader (5)	Male Leader (6)	p-value (5) vs. (6)
High outcome	52.330***	49.824***	0.667	54.947***	50.922***	0.510	55.807***	51.586***	0.550
	(3.857)	(4.414)		(4.014)	(4.655)		(4.709)	(5.318)	
Prior belief	-0.034	0.317***	0.007	0.009	0.321***	0.017	-0.018	0.334***	0.024
	(0.087)	(0.099)		(0.087)	(0.098)		(0.106)	(0.116)	
Inconsistent or non-updater	-15.549**	-4.776							
	(6.570)	(7.780)							
% endowment transferred in DG	0.409**	0.422**		0.286	0.313		0.267	0.291	
	(0.195)	(0.202)		(0.208)	(0.194)		(0.239)	(0.222)	
High Return – Low Return	0.016	-0.035*		0.016	-0.040*		0.026	-0.043*	
	(0.015)	(0.019)		(0.015)	(0.021)		(0.018)	(0.024)	
Zero return if investment fails	-2.257*	-1.413		-2.084	-1.210		-1.376	-0.987	
	(1.308)	(1.665)		(1.355)	(1.754)		(1.633)	(2.037)	
Constant	-31.634**	-30.996**		-38.085***	-29.190**		-35.412**	-21.639	
	(13.129)	(12.269)		(13.842)	(13.229)		(14.661)	(14.522)	
Outcome/Belief: Coefficient	-1524.7	157.2		6162.2	158.5		-3186.5	154.3	
Test of High outcome = 100 × Belief: p-value[(a)]	< 0.001	0.098		< 0.001	0.090		< 0.001	0.164	
Control for task order	✓	✓		✓	✓		✓	✓	
Control for Wave 2 data	✓	✓		✓	✓		✓	✓	
Control for beliefs about leader's DG behavior	✓	✓		✓	✓		✓	✓	
Excluding non-updaters				✓	✓		✓	✓	
Excluding inconsistent updaters							✓	✓	
Observations	2,000	1,930		1,880	1,800		1,450	1,450	
# participants (clusters)	200	193		188	180		145	145	
R-squared	0.253	0.217		0.248	0.213		0.244	0.212	

Robust standard errors clustered at the participant level in parentheses. Wave 2 data is the second wave of data collection conducted in 2025. We classify an evaluator as a non-updater if all their posterior beliefs are equal to their prior beliefs for all five rounds of the investment task, and as an inconsistent updater if 25% or more of their posterior beliefs are in the opposite direction to that predicted by Bayes' rule.

[(a)] This tests the null hypothesis that the coefficient on high outcome is equal to 100 times the coefficient on prior beliefs. The interpretation of the test is whether there is a difference in the *marginal change* in evaluators' discretionary payments between two scenarios: (i) with respect to a given change in outcome (from a low outcome to a high outcome), and (ii) with respect to a given change in belief (from a belief that the leader has chosen Investment Y with certainty to a belief that the leader has chosen Investment X with certainty).

*** p<0.01, ** p<0.05, * p<0.10.

Table C5: OLS regressions of positive and negative discretionary payments, by the leader's gender

	Dependent variable: Received positive discretionary payment			Dependent variable: Received negative discretionary payment		
Variables	Female Leader (1)	Male Leader (2)	p-value (1) vs. (2)	Female Leader (3)	Male Leader (4)	p-value (3) vs. (4)
High outcome	0.422***	0.297***	0.058	-0.493***	-0.329***	0.013
	(0.047)	(0.047)		(0.047)	(0.047)	
Posterior belief	0.001	0.002**	0.484	-0.001	-0.004***	0.003
	(0.001)	(0.001)		(0.001)	(0.001)	
% endowment transferred in DG	0.000	0.002		-0.002	-0.001	
	(0.002)	(0.001)		(0.002)	(0.001)	
High Return - Low Return	0.000**	-0.000		-0.000	0.000	
	(0.000)	(0.000)		(0.000)	(0.000)	
Zero return if investment fails	0.005	0.008		0.000	-0.022	
	(0.015)	(0.017)		(0.015)	(0.019)	
Constant	-0.038	0.295***		0.628***	0.699***	
	(0.091)	(0.100)		(0.101)	(0.093)	
Outcome/Belief: Coefficient	372.0	157.8		932.4	91.7	
Test of High outcome = 100 × Belief: p-value[(a)]	0.003	0.332		0.000	0.782	
Control for task order	✓	✓		✓	✓	
Control for Wave 2 data	✓	✓		✓	✓	
Control for beliefs about leader's DG behavior	✓	✓		✓	✓	
Excluding non-updaters	✓	✓		✓	✓	
Excluding inconsistent updaters	✓	✓		✓	✓	
Observations	1,450	1,450		1,450	1,450	
# participants (clusters)	145	145		145	145	
R-squared	0.250	0.170		0.319	0.282	

Robust standard errors clustered at the participant level in parentheses. Wave 2 data is the second wave of data collection conducted in 2025. This analysis excludes evaluators classified as non-updaters or inconsistent updaters. We classify an evaluator as a non-updater if all their posterior beliefs are equal to their prior beliefs for all five rounds of the investment task, and as an inconsistent updater if 25% or more of their posterior beliefs are in the opposite direction to that predicted by Bayes' rule.

[(a)] This tests the null hypothesis that the coefficient on high outcome is equal to 100 times the coefficient on posterior beliefs. The interpretation of the test is whether there is a difference in the *marginal change* in evaluators' discretionary payments between two scenarios: (i) with respect to a given change in outcome (from a low outcome to a high outcome), and (ii) with respect to a given change in belief (from a belief that the leader has chosen Investment Y with certainty to a belief that the leader has chosen Investment X with certainty).

*** p<0.01, ** p<0.05, * p<0.10.

Table C6: OLS regressions of discretionary payments, by the leader's gender (excluding evaluators who incorrectly predicted leader's gender)

	Dependent variable: Discretionary payments								
Variables	Female Leader (1)	Male Leader (2)	p-value (1) vs. (2)	Female Leader (3)	Male Leader (4)	p-value (3) vs. (4)	Female Leader (5)	Male Leader (6)	p-value (5) vs. (6)
High outcome	50.050***	40.468***	0.161	53.104***	41.078***	0.102	51.351***	36.558***	0.085
	(4.467)	(5.221)		(4.657)	(5.747)		(6.108)	(6.149)	
Posterior belief	0.103	0.374***	0.044	0.081	0.369***	0.034	0.110	0.496***	0.010
	(0.088)	(0.103)		(0.083)	(0.108)		(0.103)	(0.111)	
Inconsistent or non-updater	-15.580**	-6.373							
	(6.969)	(8.474)							
% endowment transferred in DG	0.317	0.441*		0.219	0.311		0.291	0.302	
	(0.213)	(0.224)		(0.214)	(0.216)		(0.253)	(0.242)	
High Return - Low Return	0.004	-0.025		0.010	-0.029		0.018	-0.045**	
	(0.014)	(0.016)		(0.014)	(0.018)		(0.016)	(0.020)	
Zero return if investment fails	-2.491*	-1.830		-2.255	-1.637		-1.563	-1.272	
	(1.411)	(1.577)		(1.466)	(1.664)		(1.773)	(1.924)	
Constant	-30.501**	-30.594**		-36.599**	-28.739**		-32.870**	-19.410	
	(13.839)	(13.139)		(14.723)	(14.417)		(15.532)	(15.420)	
Outcome/Belief: Coefficient	485.5	108.2		652.6	111.4		467.5	73.7	
Test of High outcome = 100 × Belief: p-value[(a)]	0.001	0.820		< 0.001	0.774		0.007	0.387	
Control for task order	✓	✓		✓	✓		✓	✓	
Control for Wave 2 data	✓	✓		✓	✓		✓	✓	
Control for beliefs about leader's DG behavior	✓	✓		✓	✓		✓	✓	
Excluding non-updaters				✓	✓		✓	✓	
Excluding inconsistent updaters							✓	✓	
Observations	1,790	1,750		1,680	1,620		1,300	1,330	
# participants (clusters)	179	175		168	162		130	133	
R-squared	0.239	0.211		0.244	0.204		0.235	0.224	

Robust standard errors clustered at the participant level in parentheses. This analysis excludes evaluators who incorrectly predicted the leader's gender in the post-experimental questionnaire. Wave 2 data is the second wave of data collection conducted in 2025. We classify an evaluator as a non-updater if all their posterior beliefs are equal to their prior beliefs for all five rounds of the investment task, and as an inconsistent updater if 25% or more of their posterior beliefs are in the opposite direction to that predicted by Bayes' rule.

[(a)] This tests the null hypothesis that the coefficient on high outcome is equal to 100 times the coefficient on posterior beliefs. The interpretation of the test is whether there is a difference in the *marginal change* in evaluators' discretionary payments between two scenarios: (i) with respect to a given change in outcome (from a low outcome to a high outcome), and (ii) with respect to a given change in belief (from a belief that the leader has chosen Investment Y with certainty to a belief that the leader has chosen Investment X with certainty).

*** p<0.01, ** p<0.05, * p<0.10.

Table C7: OLS regressions of discretionary payments, by the leader's and evaluator's gender

	Dependent variable: Discretionary payments					
	Female Evaluator			Male Evaluator		
Variables	Female Leader (1)	Male Leader (2)	p-value (1) vs. (2)	Female Leader (3)	Male Leader (4)	p-value (3) vs. (4)
(a) Full sample						
High outcome	49.219***	41.991***	0.355	48.564***	39.013***	0.355
	(4.932)	(6.168)		(7.053)	(7.724)	
Posterior belief	-0.120	0.323***	0.006	0.330**	0.396***	0.719
	(0.116)	(0.113)		(0.129)	(0.134)	
Inconsistent or non-updater	-15.970*	19.740**		-9.559	-25.054**	
	(8.529)	(9.107)		(9.424)	(11.895)	
% endowment transferred in DG	0.532	0.857**		0.263	0.337	
	(0.327)	(0.391)		(0.260)	(0.237)	
High Return - Low Return	0.045***	-0.023		-0.028	-0.015	
	(0.017)	(0.022)		(0.019)	(0.023)	
Zero return if investment fails	-0.704	-1.151		-3.405	-0.214	
	(1.513)	(2.394)		(2.167)	(2.233)	
Constant	-28.222	-43.824**		-42.192**	-30.205*	
	(20.691)	(17.253)		(17.429)	(17.731)	
Outcome/Belief: Coefficient	-410.3	129.9		147.2	98.5	
Test of High outcome = 100 × Belief: p-value[(a)]	< 0.001	0.532		0.384	0.974	
Control for task order	✓	✓		✓	✓	
Control for Wave 2 data	✓	✓		✓	✓	
Control for beliefs about leader's DG behavior	✓	✓		✓	✓	
Observations	1,010	980		980	950	
# participants (clusters)	101	98		98	95	
R-squared	0.303	0.326		0.260	0.198	
(b) Exclude non-updaters						
High outcome	50.327***	43.817***	0.431	53.385***	38.750***	0.194
	(5.208)	(6.548)		(7.498)	(8.596)	
Posterior belief	-0.048	0.265**	0.048	0.227	0.415***	0.347
	(0.099)	(0.126)		(0.139)	(0.148)	
% endowment transferred in DG	0.401	0.704*		0.192	0.251	
	(0.333)	(0.396)		(0.272)	(0.232)	
High Return - Low Return	0.046***	-0.022		-0.020	-0.022	
	(0.017)	(0.023)		(0.020)	(0.025)	
Zero return if investment fails	-0.891	-0.571		-3.116	-0.386	
	(1.616)	(2.456)		(2.184)	(2.419)	
Constant	-51.382**	-41.126**		-36.255*	-34.642*	
	(20.059)	(20.262)		(18.296)	(18.894)	
Outcome/Belief: Coefficient	-1042.6	165.4		235.0	93.3	
Test of High outcome = 100 × Belief: p-value[(a)]	< 0.001	0.308		0.115	0.893	
Control for task order	✓	✓		✓	✓	
Control for Wave 2 data	✓	✓		✓	✓	
Control for beliefs about leader's DG behavior	✓	✓		✓	✓	
Observations	940	920		930	880	
# participants (clusters)	94	92		93	88	
R-squared	0.297	0.299		0.262	0.176	

Table C7: OLS regressions of discretionary payments, by the leader's and evaluator's gender

	Dependent variable: Discretionary payments					
	Female Evaluator			Male Evaluator		
Variables	Female Leader (1)	Male Leader (2)	p-value (1) vs. (2)	Female Leader (3)	Male Leader (4)	p-value (3) vs. (4)
(c) Exclude non-updaters and inconsistent updaters						
High outcome	54.752***	34.488***	0.033	45.543***	39.244***	0.647
	(6.644)	(6.953)		(10.517)	(9.152)	
Posterior belief	-0.104	0.431***	0.002	0.392**	0.491***	0.661
	(0.113)	(0.133)		(0.188)	(0.131)	
% endowment transferred in DG	0.260	0.704		0.170	0.194	
	(0.372)	(0.502)		(0.310)	(0.257)	
High Return - Low Return	0.053***	-0.032		-0.022	-0.030	
	(0.019)	(0.028)		(0.024)	(0.026)	
Zero return if investment fails	-0.077	1.224		-2.781	-1.539	
	(1.797)	(2.906)		(2.810)	(2.830)	
Constant	-43.637**	-40.341*		-38.598*	-19.742	
	(20.106)	(21.508)		(19.815)	(20.867)	
Outcome/Belief: Coefficient	-527.6	80.0		116.2	79.9	
Test of High outcome = 100 × Belief: p-value[(a)]	< 0.001	0.629		0.818	0.600	
Control for task order	✓	✓		✓	✓	
Control for Wave 2 data	✓	✓		✓	✓	
Control for beliefs about leader's DG behavior	✓	✓		✓	✓	
Observations	740	740		700	710	
# participants (clusters)	74	74		70	71	
R-squared	0.301	0.326		0.255	0.201	

Robust standard errors clustered at the participant level in parentheses. Wave 2 data is the second wave of data collection conducted in 2025. We classify an evaluator as a non-updater if all their posterior beliefs are equal to their prior beliefs for all five rounds of the investment task, and as an inconsistent updater if 25% or more of their posterior beliefs are in the opposite direction to that predicted by Bayes' rule.

[(a)] This tests the null hypothesis that the coefficient on high outcome is equal to 100 times the coefficient on posterior beliefs. The interpretation of the test is whether there is a difference in the *marginal change* in evaluators' discretionary payments between two scenarios: (i) with respect to a given change in outcome (from a low outcome to a high outcome), and (ii) with respect to a given change in belief (from a belief that the leader has chosen Investment Y with certainty to a belief that the leader has chosen Investment X with certainty).

*** p<0.01, ** p<0.05, * p<0.10.

Appendix D Investigation of leaders' investment decisions

Table D1 reveals that leaders who transfer more in the dictator game (i.e., more altruistic) are more likely to choose Investment X. Moreover, column (5) of Table D1 reveals that leaders who anticipate a higher discretionary payment for delivering a high outcome on average are more likely to choose Investment X. Hence, consistent with our conceptual framework in Section 3, leaders' investment decisions are driven by their altruism and their beliefs about the discretionary payments they will receive.

However, columns (1) to (4) of Table D1 reveal that female leaders are also less likely to choose Investment X than male leaders.[1] This gender difference becomes smaller in column (5) when we control for the leader's expectations of discretionary payments (p-value = 0.051) and disappears in column (6) when we include interactions between the leader's gender and their expectations (p-value = 0.799). While male leaders who anticipate a higher (lower) discretionary payment for delivering a high (low) outcome are more likely to choose Investment X (p-values = 0.001 and 0.020 for anticipated payments given high and low outcomes, respectively), the interaction terms with gender suggest that female leaders are significantly less responsive than male leaders to their expectations of discretionary payments (p-values = 0.043 and 0.024 for interactions of gender with anticipated payments given high and low outcomes, respectively). Hence, this indicates that male leaders, but not female leaders, are primarily responding to their expectations about discretionary payments. Controlling for heterogeneity in leaders' responses to incentives (i.e., which is represented by λ in Appendix B) along gender lines can explain the gender difference in investment choices.

An implication of these results is that if female leaders had similar expectations about discretionary payments, they might have been as likely or more likely than male leaders to make the welfare-maximizing choice. Previous studies have also documented such self-fulfilling prophecies in discrimination, e.g., where ethnic minority workers put in less effort because they anticipate managerial biases (Glover, Pallais, and Pariente, 2017). Hence, interventions that can change expectations or societal norms about the payments provided to leaders are important for breaking such self-fulfilling prophecies and can improve decision making.

[1] Columns (1) and (2) include the decisions of all participants since all participants make decisions as leaders. Columns (3) to (6) only include participants who were eventually randomly assigned as leaders, which allows us to control for leaders' expectations of discretionary payments.

References

Glover, D., Pallais, A., Pariente, W. (2017). Discrimination as a self-fulfilling prophecy: Evidence from French grocery stores. *The Quarterly Journal of Economics,* 132(3):1219-1260.

Table D1: OLS regressions of leaders' investment choice

	Dependent variable: = 1 if leader chooses high investment					
	All participants		Participants assigned as leaders			
Variables	(1)	(2)	(3)	(4)	(5)	(6)
Female leader	-0.057**	-0.050*	-0.110**	-0.111**	-0.095*	0.016
	(0.027)	(0.027)	(0.050)	(0.048)	(0.048)	(0.062)
% endowment transferred in DG	0.003***	0.003***	0.002**	0.002*	0.002*	0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
# risky choices in RT	-0.005	-0.006	-0.019	-0.024*	-0.026*	-0.028**
	(0.008)	(0.008)	(0.015)	(0.014)	(0.014)	(0.014)
High Return – Low Return	0.002***	0.002***	0.002***	0.002***	0.002***	0.002***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Zero return if investment fails	0.064***	0.064***	0.038	0.038	0.038	0.039
	(0.017)	(0.017)	(0.027)	(0.027)	(0.027)	(0.027)
Expected payment for high outcome					0.001***	0.002***
					(0.000)	(0.001)
Female leader × Expected payment for high outcome						-0.002**
						(0.001)
Expected payment for low outcome					-0.000	-0.001**
					(0.000)	(0.001)
Female leader × Expected payment for low outcome						0.002**
						(0.001)
Constant	-0.110*	-0.247**	0.125	0.005	-0.003	-0.119
	(0.064)	(0.107)	(0.112)	(0.205)	(0.209)	(0.199)
Control for task order	✓	✓	✓	✓	✓	✓
Control for Wave 2 data	✓	✓	✓	✓	✓	✓
Individual controls		✓		✓	✓	✓
Observations	2,950	2,950	985	985	985	985
# participants (clusters)	590	590	197	197	197	197
R-squared	0.047	0.061	0.050	0.097	0.110	0.123

Robust standard errors clustered at the participant level in parentheses. DG: Dictator Game; RT: Risk Task.

Wave 2 data is the second wave of data collection conducted in 2025. We classify an evaluator as a non-updater if all their posterior beliefs are equal to their prior beliefs for all five rounds of the investment task, and as an inconsistent updater if 25% or more of their posterior beliefs are in the opposite direction to that predicted by Bayes' rule. Note that all participants make investment decisions as leaders in the experiment. Columns (1)-(2) include data from all participants, while columns (3)-(5) include data from those participants who were assigned the roles of leader during the experiment. In columns (2), (4), (5), and (6) we also control for participants' characteristics, which include their age, whether the participant is pursuing a major in economics, whether the participant is an undergraduate student, whether the participant is Australian, previous experience with economics experiments, and CRT score. Probit regressions of leaders' investment decisions provide similar conclusions.

*** p<0.01, ** p<0.05, * p<0.10.